MATH 3650-002 Final Project: Processes for Image Deblurring

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Abstract

Image deblurring is a common occurrence in the modern day. Modern phones take a flurry of images and combine them along with debluring algorithms in order to create a final picture free from camera movement or shaky hands. This paper will analyze the process, and attempt to replicate the simple procedures using matlab. While the image deblurring is the main focus of the report, the math behind the algorithms will also be highlighted. The methods used for this project are Tikhonov Regularization, Wiener Filtering, and Truncated Single Value Decomposition. These three methods were applied to a given set of images, and yielded differing results. With the usage of these methods, the images were deblurred to differing extents.

Introduction

In today's digital age, our society captures and views an immense amount of digital images every day. While improvements in technology continue to enhance image quality, forms of degradation like camera motion and lack of focus persist in many photos taken. This ability to recover clear visual data is not only valuable in consumer photography; it also plays a critical role in fields like astronomy and medicine, where accurate imaging is essential, and the smallest details make large differences. Image deblurring is therefore, an extremely important topic to examine. This project studies three deblurring techniques, observing how each addresses the issues faced when reconstructing an image with limited data. Through evaluation of each technique's effectiveness in enhancing stability, noise sensitivity, and overall quality, a deeper understanding as to what factors positively influence image deblurring can be developed.

Methods and Models

Tikhonov Regularization

The first method used in this project is the Tikhonov Regularization. Named due to his work with systems of equations, where small measurements would cause huge changes in solutions. He proposed an additional term that would add stability by penalizing the large magnitudes. This can be modified using a constant, and in terms of image deblurring, these can

control the balance between smoothness and detail. In terms of the project, the Tikhonov parameter will be represented as α. The parameter will be solved using the following equation.

$$G(\alpha) = \frac{\sum_{i=1}^{N} \left(\frac{u_i^T b}{\sigma_i^2 + \alpha^2}\right)^2}{\left(\sum_{i=1}^{N} \frac{1}{\sigma_i^2 + \alpha^2}\right)2} \quad (1)$$

α will then be used for generalized cross-validation (gcv). Tikhonov's regularization begins with converting the image to a double matrix. This matrix stores all values on a scale with values ranging from 0-1 (white to black). For Tikhonov's method to apply, the blur is assumed to be consistent. The provided code to find the PSF and its center is utilized to generate both. However, the PSF matrix is smaller than the image matrix and is supplemented with a zero matrix, so the sizes are the same. A fast Fourier transformation (FFT) is taken of both the image and the circular shift of the PSF matrix. This results in functions that represent the values, as opposed to numerical values. These functions are able to be converted back into their numerical values and preserve the original quality. The filtering applied to this image is periodic filtering. While reflective filtering is the most accurate for images with complex detail, the provided functions for utilizing Tikhonov regularization are better suited for periodic boundary conditions. Now that there is a matrix for the image and one for the filter, an α value will be found to ensure a balance between the smoothness and detail. This is normally done by using the GCV function to solve for α, as seen above in Equation 1. This α value is the distortion, and will be factored into the zero matrix before the final filtering operation, which can be seen below.

$$\widehat{X} = \frac{\widehat{B}}{\bar{S} + |\alpha|^2} \quad (2)$$

This equation divides the FFT of the original image $\widehat{B}$, by the sum of the conjugate of the FFT of the circularly shifted filter matrix $\bar{S}$ and the square of the factor of distortion α. The final step for deblurring an image is to take the inverse fast-Fourier transformation of the Tikhonov value $\widehat{X}$. The original and final images can be seen in Figure 1.

Wiener Filter

The second method used is the Wiener filter. This method repeats a few steps from the first, those being the creation of a matrix representation of the image, the constant blur assumption, and the PSF applied to a zero matrix the size of the image matrix. To progress the Wiener filter, a circular shift is placed over the zero matrix. This is done because the Fast Fourier Transform (FFT) applied next assumes the numbers to be shifted circularly. The FFT outputs complex numbers that use a combination of sine and cosine functions, represented as $(\widehat{I})$. A Weiner parameter k is needed in order to adjust for noise. This value is estimated through trial and error. Finally, the FFT is taken of the double of the image matrix, resulting in $(\widehat{C})$. This double is taken so that there are values over the boundary for the deblur to apply to. These will be cropped out, but make the edges as smooth as the rest of the final image. With the FFT completed, the Weiner parameter estimated, and the FFT of the double of the image matrix, these can be combined for the next step.

$$\widehat{W} = \frac{conjugate(\widehat{I})}{(\left|\widehat{I}\right|^2 + k) \cdot \widehat{C}} \tag{3}$$

The resultant matrix $\widehat{W}$ is still written in frequencies and requires an inverse FFT to return to its original state, now filtered.

Truncated Single Value Decomposition

The final method utilized for deblurring images was the truncated single value decomposition (TSVD). Like the previous two methods TSVD has the same base steps: converting the image into a double-matrix, and gathering the PSF with the given function. The next steps are where TSVD differs from the previous methods. A tolerance is estimated using standard conventions, which stipulate a number around 0.01, but it can be larger or smaller. Then, a circular shift of the PSF with respect to its center is performed, followed by an FFT of said circular shift. Then a second FFT is calculated from the double-matrix of the original image. Next, a binary mask ϕ is found by getting the absolute value of the circularly shifted FFT that is greater than the tolerance. ϕ is converted into an index set, which is true anywhere an inversion

is allowed. Then, the following equation is performed to solve for the final TSVD matrix:

$$\widehat{X} = \widehat{B} \bullet \frac{\phi_i}{S_i} \tag{4}$$

The variables used in the TSVD equation are as follows: $\widehat{B}$ is the FFT of the double-matrix of the original image, ϕ_i is the binary max of the index value, i, lastly is S_i which is the FFT of the circular shift of the PSF, with respect to the index value.

Results and Discussion

The first image that was deblurred was ‘successkid_blur.jpg’ (Figure 1), which appeared to have the heaviest level of blur applied to it of any of the images. We chose to utilize the Tikhonov filter in this case because we knew the final image would have fine details, and the Tikhonov was well-suited for recovering those. However, the final result (Figure 1) was not completely restored; this can be attributed to an ill-chosen alpha value. While a GCV could be performed to find the ‘best’ alpha, the provided functions resulted in blank images.

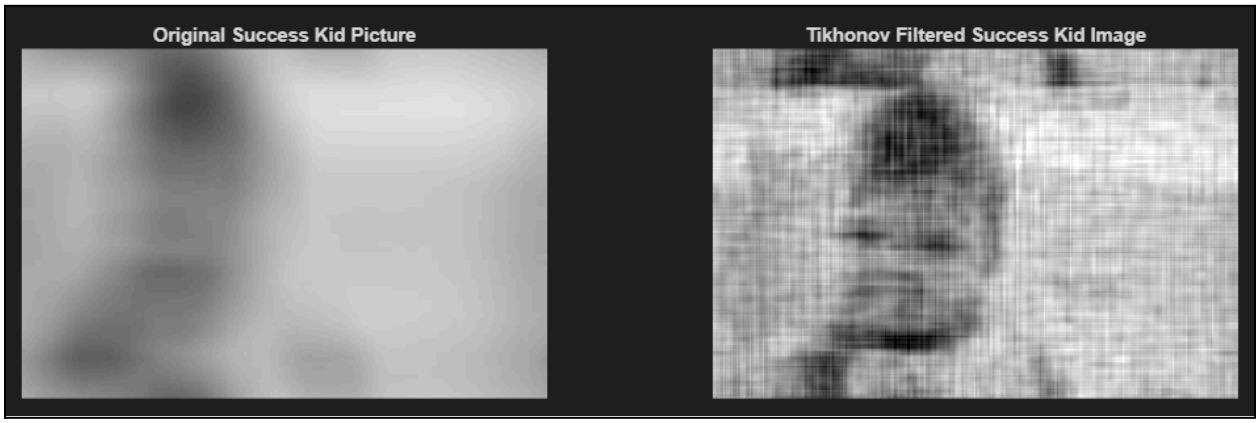


Figure 1 - Tikhonov Regularization of ‘successkid_blur.jpg’ and original

The second image deblurred was ‘batman_blur.jpg’ (Figure 2). For this image, we chose to use Wiener filtering. The de-blurred Batman image (Figure 2) had very little loss of detail, and high-quality restoration. The only artifacts are small, and there is a slight distortion at the border as a result of using a periodic boundary condition. The periodic boundary condition

continues the image generation using values from the opposite side. The original image has differences in color on opposite sides. These can be seen on the filtered image.

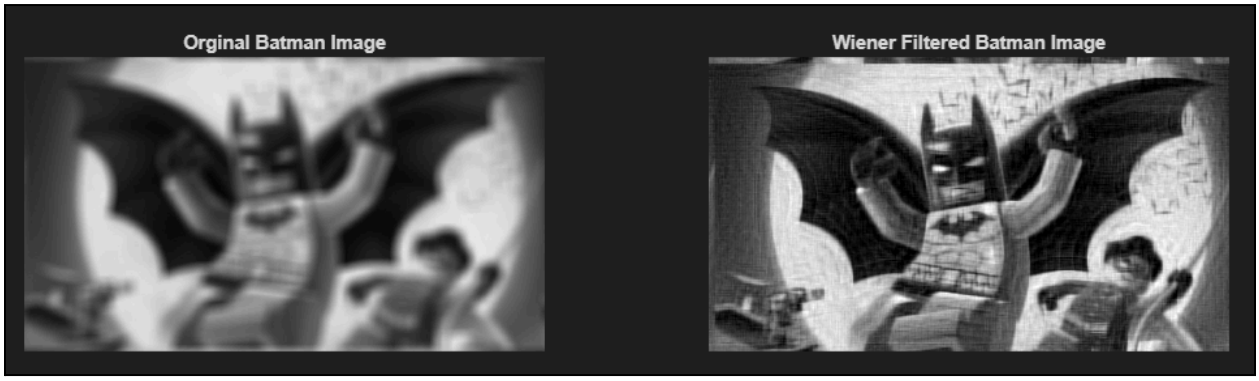


Figure 2 - Wiener Filtered ‘batman_blur.jpg’ and original

The third image deblurred was ‘panda_blur.jpg’ (Figure 3). As the other two methods were already used, TVSD was implemented. Looking at the artifacts on the deblurred image, (Figure 3), there is a high level of distortion. This is because TSVD is best for very simple images, so high detail is prone to having a high amount of artifacts.

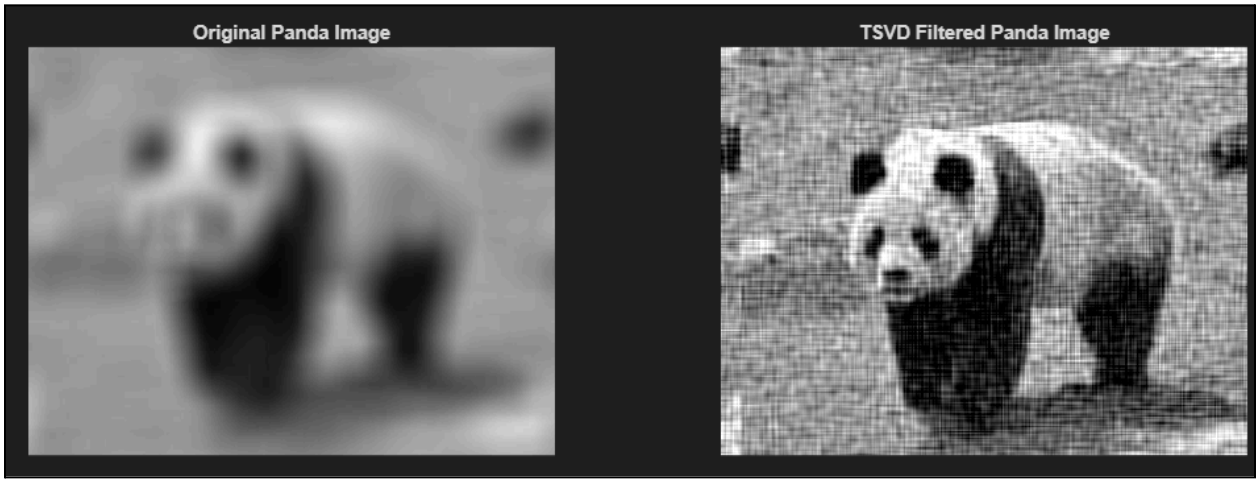


Figure 3 - TSVD Filtered ‘panda_blur.jpg’ and original

The fourth image deblurred was ‘GrumpyCatThumb_blur.jpg’ (Figure 4). Wiener filtering was chosen over Tikhonov and TSVD because Wiener is better for high detail images, and is quicker to compute than Tikhonov. Looking at the deblurred cat image (Figure 4), there is little distortion or artifacts on the filtered image. This shows the efficacy of the Wiener method.

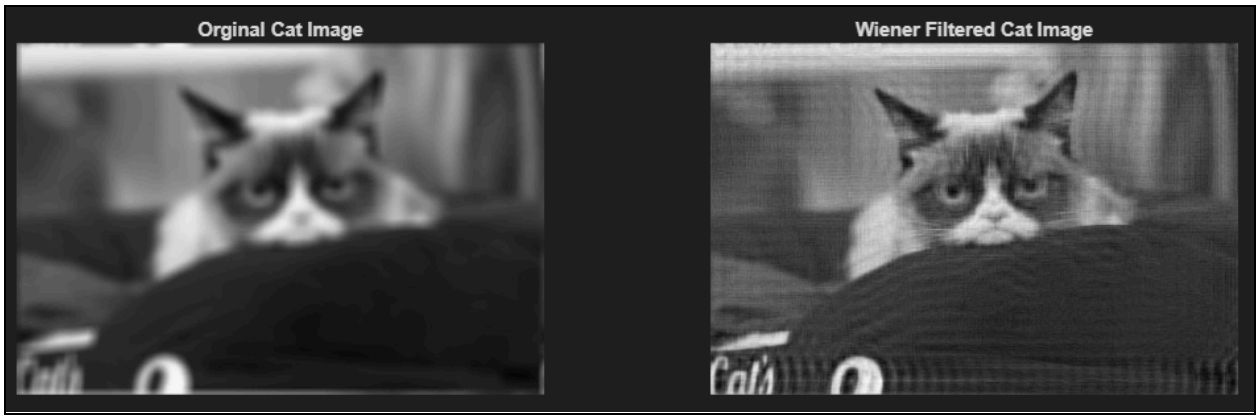


Figure 4 - Wiener Filtered ‘GrumpyCatThumb_blur.jpg’ and original

The final image ‘raccoon_blur.jpg’ (Figure 5) was deblurred with the TSVD method because the raccoon in the foreground had a strong contrast to the background ,allowing for a clear subject. However, examining the actual deblurred image (Figure 5), it is apparent that the background was too detailed to produce a clear result, but the raccoon is still identifiable.

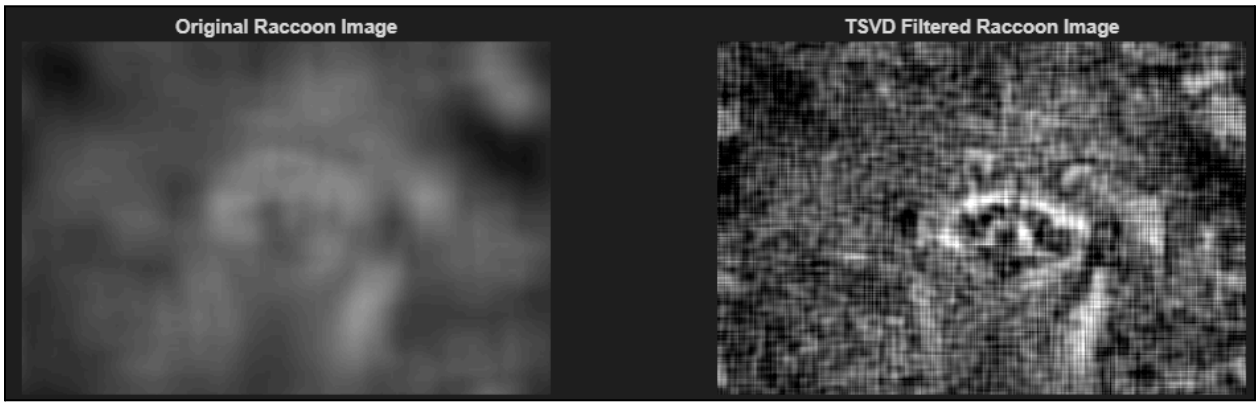


Figure 5 - TSVD Filtered ‘raccoon_blur.jpg’ and original

Three mathematical approaches to image deblurring were used. Tikhonov Regularization, Wiener Filtering, and TSVD all demonstrated their differences in performance using images with varying blur characteristics. The results showed that none of the methods are universally optimal, as compromises are made in detail preservation, computation time, and artifact mitigation. The TSVD images reinforced that the method is best for less complex images, while Wiener filtering proved its efficacy with highly detailed images. Tikhonov Regularization finds itself balancing between the two. These differences show how each method’s variation in its

construction and assumptions lead to different results, deepening our understanding of how mathematical techniques influence the reconstruction of visual data.

References

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